Assel Saginayeva AIB 2602

AI Business Model Stress Test

This report examines how Ferrero, one of the world's leading confectionery companies, applies artificial intelligence across its business. Rather than providing a comprehensive overview of Ferrero's operations, the analysis is deliberately scoped to its AI initiatives alone. For each AI application identified, the report asks a single guiding question: should Bayan Sulu adopt a similar approach? The aim is to translate Ferrero's experience into practical, actionable insight for Bayan Sulu's own path toward AI adoption.

We sort each idea into one of five groups:

- **Adopt** — Do it almost the same way Ferrero does. Bayan Sulu is ready for this.
- **Adapt** — Use the idea, but make it smaller or simpler to fit Bayan Sulu.
- **Pilot** — Try it on a small scale first, like one product or one city.
- **Delay** — Wait. This could make sense later, but not right now.
- **Reject** — This does not fit Bayan Sulu. The cost or the need is not there.

Ferrero: areas where AI is used

Here are six ways AI use in Ferrero's business:

1. **Predicting what stores need.** In the United States, Ferrero uses a tool called Alloy to track store sales data in real time. This helps Ferrero guess when a store is about to run out of a product, or when a product looks like it's on the shelf but is actually gone. This has saved Ferrero more than 5% a year by cutting waste and missed sales [1].
2. **Tracking where ingredients come from (confirmed).** Ferrero uses two tools, called Sourcemap and Starling, to trace its ingredients back to the farm. Today, Ferrero can trace over 90% of its cocoa, palm oil, and hazelnuts this way [2].
3. **Watching farms by satellite (confirmed).** Ferrero uses satellites, plus AI that reads the images, to check its palm oil farms for illegal tree-cutting. Ferrero has promised to watch 100% of its palm oil farms this way [3].
4. **Learning about customers and showing different ads (confirmed).** Ferrero uses a Google tool called BigQuery to study visitors on its apps and websites. It groups visitors by what they like, then shows each group different ads [4].
5. **Using AI to design 7 million different jar labels (confirmed).** For a Nutella campaign in Italy, Ferrero used a computer program to design 7 million different jar labels, no two exactly the same. The jars sold out [5].
6. **Using AI cameras to sort nuts (common in this industry, not confirmed for Ferrero).** Companies that process hazelnuts at Ferrero's size often use AI-powered cameras to sort good nuts from bad ones. We could not find a news story saying Ferrero itself does this, but it fits a company of Ferrero's size [6].

Bayan Sulu: Where It Stands Today

Bayan Sulu is the biggest candy and chocolate maker in Kazakhstan. It started in 1974 and is based in the city of Kostanay [7]. It is a much smaller company than Ferrero.

- **Size:** Bayan Sulu makes more than 300 different products. It can make 80,000–85,000 tons of food a year. It earns about 20.5 billion Kazakhstani tenge a year — around $48 million. It has more than 1,500 workers, and anyone can buy its stock on the Kazakhstan Stock Exchange [8].
- **Where it sells:** About 40% of what Bayan Sulu makes is sold in 14 other countries, including China, Mongolia, South Korea, and nearby Central Asian countries. But at home in Kazakhstan, imported candy still takes up more than 55% of the market. In richer countries, imports usually take up only about 5% [8].
- **A real problem it already faced:** In 2022, a sugar shortage showed that Bayan Sulu depends too much on imported sugar and does not keep a big enough backup supply [8].
- **How it spends money:** Over the last 15 years, Bayan Sulu has spent more than $100 million on new factory equipment. It is also planning to spend about 13 billion tenge (around $29 million) on a new factory. Bayan Sulu has earned safety and quality certificates (ISO 9001, ISO 22000, ISO 50001), a Halal certificate, and an award for treating men and women fairly at work (EDGE). There is no exact public sign that it has spent money on AI or data tools [8].
- **How ready it is for AI:** There is no public proof that Bayan Sulu uses AI, big data tools, or a customer-tracking system today. This is a real gap compared to Ferrero. But it also means Bayan Sulu can start fresh — it has no old AI system to replace, because it has none yet.

The Stress Test

AI idea	What Ferrero does	Our call for Bayan Sulu	Why
Predict what stores need	Tracks real store sales data in the U.S. to avoid running out of stock or having too much [1]	**Pilot**	Kazakhstan stores are a small, safe place to test this before trying it in 14 different export countries
Trace ingredients back to the farm	Traces over 90% of its cocoa, palm oil, and hazelnuts [2]	**Adapt**	Bayan Sulu's real risk is smaller — mainly imported sugar and African cocoa. A simple tracking tool aimed at the 2022 sugar problem makes more sense

AI idea	What Ferrero does	Our call for Bayan Sulu	Why
Watch farms by satellite for illegal tree-cutting	Watches 100% of its palm oil farms by satellite [3]	**Delay**	Bayan Sulu does not buy much palm oil and sells mostly to markets that don't require this yet.
Study customers and show different ads	Groups app and website visitors, then shows each group different ads [4]	**Pilot**	Bayan Sulu's website and app have far fewer visitors than Ferrero's. Test this on one channel before building a big system
Use AI to design many unique product labels	Designed 7 million unique Nutella jar labels; they sold out [5]	**Adapt**	A smaller, one-season version on one product fits Bayan Sulu's budget better than a big national campaign
Use AI cameras to sort ingredients	Common in this industry for companies Ferrero's size [6]	**Adopt**	This is an off-the-shelf tool, not a new invention. It fits how Bayan Sulu already spends money on new equipment, and it would cut waste right away

Roadmap

- **First year:** Start using AI cameras to sort ingredients on the line that wastes the most. Test the "predict what stores need" idea with one or two stores in Kazakhstan.
- **Year two:** Build a simple tool to track sugar and cocoa supply risks. Test the customer-and-ads idea on just one channel, like the website or a loyalty app.
- **Year three and later, if earlier steps work:** Try a smaller version of the "many unique labels" idea on one holiday product.

References

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